

nrphy: An Open-Source Simulation of a 5G NR OFDM Transceiver

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Abstract—Every wireless engineer knows the OFDM block diagram: bits through FEC, interleaving and QAM onto a pilot-bearing grid, IFFT, cyclic prefix and windowing into a DAC, an I/Q modulator and a power amplifier, then a time-varying multipath channel, and a receiver that synchronises, estimates, equalizes one tap per subcarrier and decodes. This paper presents *nrphy*, a pure-numpy, fully vectorised simulation of that diagram built to the 5G NR standard wherever the standard defines bits: the TS 38.212 LDPC with its circular-buffer rate matching and modulation interleaver, CRC24A block verdicts, the TS 38.211 constellations, Gold sequences and type-1 DM-RS, normal-CP numerology that lands a 14-symbol slot on exactly half a millisecond, and TR 38.901 TDL fading profiles with Jakes Doppler. The receiver is blind: timing and fractional CFO come from the cyclic prefix, integer CFO and absolute slot identity from the DM-RS through a ramp-invariant differential metric, the channel from least squares with a ramp-aware DFT denoiser, and noise variance from pilot residuals. What the paper adds to the usual waterfall is a decibel budget. At rate 1/2 the coded cliffs land 1.4 dB from constrained capacity for both QPSK and 16QAM, and the package splits that gap into about 1.0 dB of min-sum decoding, 0.4 dB of blind synchronisation and estimation, and under 0.1 dB of front ends. A matching budget splits transmit EVM by impairment, and its shares add back to both measured totals exactly. With clean analog at 30 dB the chain averages 3.1–3.5% EVM against a 3.16% thermal floor, so the receiver itself is transparent. The package is the third of a family, after coherent fibre and free-space siblings, and is released for others to complete.

Index Terms—5G NR, OFDM, LDPC codes, channel estimation, synchronisation, open-source software, simulation.

I. INTRODUCTION

THE OFDM transceiver diagram is the most-drawn figure in wireless engineering, and 5G NR is its largest deployment. The standard’s physical layer is public and defined to the bit in TS 38.211 and TS 38.212 [1], [2], its channel models in TR 38.901 [3], its transmit quality floors in TS 38.101 [4]. What a student or an engineer starting a receiver project wants is the whole diagram in one readable place: every block a page of numpy, every standard structure bit-exact, every measurement reproducible. Research platforms exist [5], [6], and they are excellent at what they are for (GPU-scale link-level studies, running networks); what they are not is small.

nrphy is that small executable diagram, the third of a family built to one discipline: the author’s *ofmphy* package simulates the 800G coherent fibre interface and *fsophy* the SDA-OCT free-space link, and all three share the same rules. Standard structures are implemented to the letter and unit-tested against the specification’s numbers; the receiver earns its numbers blind; and validation is anchored on exact theory on one side and the standard’s own limits on the other. The LDPC here is literally the same code the free-space package borrowed from

5G, now back in its native habitat with the standard’s full rate matching around it.

One habit this paper argues for deserves naming up front. A coded waterfall on its own says little: a cliff 1.4 dB from capacity could be a good code with a sloppy receiver or a mediocre code with a perfect one, and the plot cannot tell you which. Because every block here is a switch, the same transport block can be run through the code alone, then through the whole waveform with a blind receiver, then with the front ends on, and the threshold measured at each rung. What comes out is a budget rather than a number, and it says where the effort belongs. The same trick on EVM turns one percentage into a list of named culprits that adds up. The contributions are:

- the complete OFDM transceiver of the block diagram, in pure numpy, one readable module per stage;
- bit-exact 38.212 coding (BG1 LDPC, CRC24A, circular-buffer rate matching with repetition, the Q_m -row interleaver) and bit-exact 38.211 modulation, numerology, Gold sequences and type-1 DM-RS, with unit tests pinning the exact constellation points and slot timings;
- a blind receiver whose synchronisation and estimation machinery is built ramp-invariant, for reasons the paper explains rather than hides; and
- validation by decomposition: a decibel budget from capacity to the full chain, an EVM budget by impairment, exact uncoded QAM theory, and the 38.101 limits.

The library is meant to grow by contribution: Section VIII lists what it does not yet have.

II. RELATED WORK

Sionna [5] is the research standard for differentiable link-level simulation and implements far more of NR than this package ever will; OpenAirInterface [6] runs real networks; MATLAB’s toolboxes are complete but neither free nor inspectable. *nrphy* sits where its siblings sit: not a platform, not a product, but a readable executable form of the standard chain, small enough that every block can be read in one sitting and checked against the specification it implements.

III. TRANSMITTER

Fig. 1 shows the chain, drawn as the block diagram it implements. One transport block is one code block of the largest BG1 size: $A = 8424$ payload bits plus CRC24A fill the information block $K = 8448$ exactly, and blocks follow each other across slots for as long as the run asks.

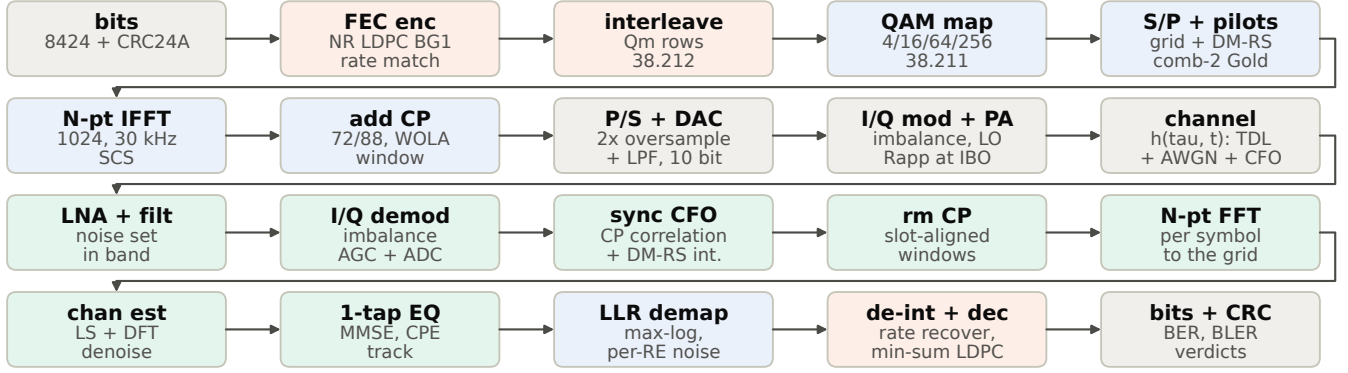


Fig. 1. System overview, block for block the diagram of the title: transmit path, front ends, the fading channel, and the blind receiver with its decoders.

A. Coding, rate matching, and the interleaver

The FEC is the NR LDPC, base graph 1 at lifting $Z = 384$. The 46×68 base matrix is a table of shifts, and each entry expands into a $Z \times Z$ circulant permutation, so an edge of the protograph is one `np.roll` of a length- Z vector and the whole decoder vectorises without a single loop over bits. Encoding solves the four core parity groups through the double-diagonal structure the base graph was designed around, then reads the 42 extension parities straight off their rows, one row at a time, because each of those rows touches exactly one new parity block.

Rate matching is where NR does something that looks wrong at first sight. The circular buffer holds the codeword minus its first two systematic blocks, $N_{cb} = 66Z$, so the transmitter throws away $2Z = 768$ information bits before it sends anything. Those punctured bits are not lost: they are the most heavily connected variable nodes in BG1, the decoder recovers them from their many edges, and leaving them out of the buffer is what lets the same base graph reach rate 8/9. The buffer is read cyclically, so below rate 1/3, where E runs past N_{cb} , bits repeat and combine as summed LLRs. Then the bits are interleaved across modulation levels, $f_{iQ_m+k} = e_{kE/Q_m+i}$, which is a Q_m -row write-column-read permutation. That interleaver is the reason a deep fade or a weak constellation level does not concentrate on one stretch of the code: in 16QAM the outer PAM level is markedly less reliable than the inner one, and without the interleaver a contiguous run of code bits would inherit that weakness together. A Gold-sequence scrambler (clause 7.3.1.1 shape) whitens the bits last, so the waveform statistics never see the code's structure and two users on the same resources do not correlate.

B. The grid and its pilots

Numerology is 30 kHz subcarrier spacing, 51 resource blocks (612 subcarriers) on a 1024-point FFT at 30.72 Msps: a 20 MHz carrier. The cyclic prefix arithmetic is best done once by hand, because it is the one place where the standard's numbers look arbitrary and are not. At $\mu = 1$ the prefix is $144 \cdot 2^{-\mu}$ samples at the reference rate, which scales to 72 samples here, and $14 \times (72 + 1024) = 15344$, sixteen samples short of the 15360 that half a millisecond takes at 30.72 Msps.

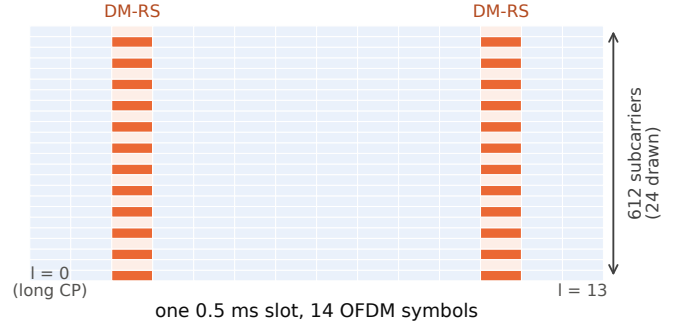


Fig. 2. One slot of the resource grid: type-1 DM-RS combs on symbols 2 and 11 (data on their odd subcarriers), data elsewhere, long cyclic prefix on symbol 0.

The standard puts those sixteen samples on the first prefix of the slot, making it 88, and the slot closes exactly. That single irregularity is invisible in the transmitter and, as Section V shows, is the difference between a synchroniser that holds and one that walks.

DM-RS is configuration type 1 on symbols 2 and 11: QPSK pilots on every second subcarrier, drawn from the length-31 Gold sequence with the standard's $c_{\text{init}}(N_{\text{ID}}, \text{slot}, l)$. Because the seed carries the slot number and the symbol index, no two pilot symbols in a frame look alike, and a receiver that finds a match learns not just that it is looking at a pilot but at which pilot. Slot identity in this package is recovered from exactly that property. Comb-2 also means the pilot symbols still carry data on their odd subcarriers, so the pilot overhead is 612 resource elements out of 8568 in a slot, about 7%. Fig. 2 draws one slot.

C. Waveform: IFFT, cyclic prefix, windowing

Each symbol is an IFFT of its subcarriers, prefixed cyclically. The cyclic prefix is what buys OFDM its one-tap equalizer, and the mechanism is easy to state in full: a channel of length L convolves linearly with the transmitted block, but if the last L samples of the block are pasted in front of it, the receiver's FFT window sees what a *circular* convolution would have produced, and circular convolution diagonalises in the DFT basis. Every subcarrier then carries one complex

gain and nothing else. The price is the prefix itself, 72/1096 of the airtime here, about 6.6%.

Windowing (WOLA) tapers W samples of each symbol edge with a raised cosine and overlap-adds neighbours. A plain CP changes the waveform abruptly at every boundary, and an abrupt change in time is a slow decay in frequency, so the spectrum falls off far more gently than the occupied bandwidth suggests. The taper smooths those joins. What it costs is not time but prefix margin: the tapered samples are taken from the head of the prefix, which was redundant anyway, so the symbol rate does not change and the only real loss is W samples of the delay-spread budget. Fig. 4b shows what that buys.

D. DAC, I/Q modulator, PA

The waveform is oversampled twice for the analog stages, quantised to 10 bits, given the I/Q modulator's gain and phase imbalance (0.2 dB and 1°) and a -35 dBc LO leakage, and pushed through a Rapp solid-state PA model [7] at a configured input backoff. This is where OFDM's famous peak-to-average power ratio stops being folklore and starts being EVM. A block of 612 independently modulated subcarriers adds up to something very close to complex Gaussian by the central limit theorem, and a complex Gaussian visits large peaks often enough to matter: the CCDF of Fig. 4a crosses 10^{-2} at 10.7 dB and 10^{-3} at 11.5 dB. An amplifier sized for the average power therefore spends a measurable fraction of its life in compression, and Table II prices exactly what that costs.

IV. THE CHANNEL: $h(\tau, t) + \text{AWGN}$

The channel is the tap-delay line of TR 38.901: the TDL-A/B/C profiles (NLOS Rayleigh) and TDL-D/E (Rician first tap) with every normalised delay scaled by the configured RMS delay spread, each tap fading with a Jakes Doppler spectrum built from a sum of sinusoids [8], and fractional tap delays honoured on the sample grid through windowed-sinc interpolation. A carrier frequency offset (1.4 subcarrier spacings by default, so the integer and fractional parts both matter) rotates the whole waveform. Noise is added in-band at the base rate against measured signal power, so the configured E_s/N_0 per occupied subcarrier is exact by construction rather than tracked through the chain's scalings.

The default fading case needs one caveat stated plainly. TDL-C scaled to 300 ns RMS delay spread has its last tap near $2.6 \mu\text{s}$, while the normal cyclic prefix at 30 kHz spacing is $2.34 \mu\text{s}$, so the tail of the profile falls outside the prefix and contributes a little interference that no equalizer removes. This is not a modelling slip; TDLC300 is itself one of the propagation conditions of TS 38.101-4, and the standard pairs it with these numerologies knowingly, because the taps out there carry very little energy. The effect is visible as a shallow error floor in the pre-FEC curve of Fig. 6 rather than as a failure, and the honest reading is that the last few tenths of a decibel at high SNR on that curve are the profile, not the receiver.

V. RECEIVER

The receiver is blind: timing, CFO, slot identity, channel and noise variance are estimated from the waveform and the DM-RS. The known transmit symbols are used to measure EVM, never to receive. The design follows the structure Speth *et al.* laid out for broadband OFDM receivers [9]: acquire coarsely from the waveform's own redundancy, refine on the pilots, and keep a per-symbol phase correction running underneath everything else.

A. Synchronisation

Timing and fractional CFO come from the cyclic prefix itself, van de Beek's ML estimator [10]: every OFDM symbol repeats its own tail, so $r[n] r^*[n+N]$ is coherent for as long as the correlation window stays inside a prefix. Its magnitude says where the symbol starts, and its phase says how far the carrier has drifted, since N samples of delay is exactly one subcarrier spacing of phase. Three details separate an estimator that works from one that only appears to.

The first is the fold. Stacking the correlation from many symbols is what makes the peak stand out of the noise, and the obvious way to stack is to fold the correlation at the symbol period. That is wrong here, and wrong in a way that hides itself: the first prefix of each slot is sixteen samples longer than the others, so a symbol-period fold slips by sixteen samples every slot and the peak walks late in proportion to how long the capture is. A short test passes and a long run degrades. The fix is to fold at the slot period instead, summing the fourteen prefixes at their true offsets, which costs nothing and is correct for any capture length.

The second is the taper. WOLA has already smoothed the head of every prefix, so the correlation over those samples is neither the symbol's own tail nor the neighbour's, and including them biases the peak toward the taper. The correlation window therefore starts after the windowed region and runs only over the clean part of the prefix.

The third is the sign of the error. Having found the peak, the estimate is deliberately backed off into the prefix, because the two directions are not symmetric. Sampling early leaves the FFT window entirely inside the prefix, which is a cyclic shift of the symbol, which is a pure phase ramp across subcarriers, and the channel estimator absorbs it without complaint. Sampling late drags the next symbol into the window, and that is intersymbol interference, which nothing absorbs. Half the clean prefix is the margin used here, which is also roughly what a channel's own delay spread wants. Measured over six seeds with every impairment on, the estimate lands 16 to 34 samples early inside a 72-sample prefix, and that spread does not grow as the capture goes from one transport block to sixteen.

The prefix says where a symbol starts but not which symbol it is, and its phase only resolves the carrier offset inside one subcarrier spacing. The DM-RS settles both. With $v_k = r_k \text{ref}_k^*$ at each pilot, the product $v_k v_{k+1}^*$ has a phase that does not depend on the timing ramp at all, so summing it stays coherent across all 306 pilots where a plain correlation would wind itself to zero. The search runs over frequency shifts and

over all four candidate references (two pilot symbols, two slot parities), because thirteen of the fourteen fold offsets still line up when the prefix correlator locks one symbol over, and a receiver that assumes it found symbol 2 of slot 0 will happily decode slot 1 with slot 0's pilots and produce nonsense. Searching both parities and returning the offset back to absolute slot 0 removes the failure entirely. Residual CFO after all this is under 30 Hz of the 41 kHz applied, a thousandth of a subcarrier spacing.

B. Channel estimation

Least squares at the pilot comb, then a DFT denoiser. The true channel lives inside the CP delay, so all the LS estimate shows beyond it is noise, and zeroing that keeps all the signal while dropping most of the noise: 306 pilot bins collapse to the 50 that can hold a prefix-length channel, near 8 dB of estimation SNR for free. The one subtlety is again the ramp. The timing backoff shifts the delay-domain impulse away from zero, and a brickwall window applied to the raw ramped estimate cuts through the middle of the impulse and rings at the band edges, which was measured to cost four times in EVM before it was found. The denoiser therefore estimates the linear phase slope from neighbouring pilots, derotates it, windows around zero delay with tapered rather than square edges, and re-applies the slope. Odd subcarriers interpolate linearly across the comb, symbols between DM-RS positions interpolate linearly across time, and the pilot residuals against the final estimate give the per-RE noise variance with no genie anywhere.

C. Equalization and demapping

The equalizer is the one-tap MMSE the cyclic prefix promised, $\hat{X}_k = H_k^* Y_k / (|H_k|^2 + \sigma^2)$, with the MMSE bias divided back out so the constellation is not shrunk toward the origin at low SNR, and a decision-directed common-phase correction per symbol absorbing what residual CFO leaves. LLRs are max-log per PAM axis, which is exact rather than a shortcut here because the 38.211 constellations factor into independent I and Q PAM sets by construction. Each LLR is weighted by its own resource element's post-equalizer noise variance $\sigma^2 / |H_k|^2$, because after a fading channel every subcarrier lives at its own SNR and a decoder that is told which bits came from the faded subcarriers does markedly better than one that is not. Descrambling, de-interleaving and rate recovery (summing repeated LLRs, zeros for the punctured systematic blocks) hand the circular buffer to the normalised min-sum decoder at $\alpha = 0.75$, with syndrome-based early termination, and CRC24A delivers the standard's own block verdict.

VI. VALIDATION AND RESULTS

A. The decibel budget

Table I is the measurement this package exists to make. The same transport block is taken through four settings and the E_s/N_0 for 10% BLER, the 3GPP link-level operating point, is found by bisection in each. The first row is the BICM mutual

TABLE I
WHERE THE DECIBELS GO. E_s/N_0 FOR 10% BLER AT RATE 1/2, AND
WHAT EACH STAGE ADDS OVER THE ONE ABOVE IT.

Stage	QPSK		16QAM	
	dB	adds	dB	adds
BICM capacity	0.19		5.28	
+ LDPC, exact LLRs	1.18	0.99	6.22	0.94
+ OFDM, blind Rx	1.61	0.43	6.64	0.42
+ DAC, I/Q, PA, CFO	1.61	0.00	6.71	0.07

information of the constellation, computed numerically by the package itself, which is the most any receiver could do with these bits on this alphabet. The second adds the real code with exact LLRs and no OFDM at all. The third wraps the whole waveform and the blind receiver around it with ideal analog. The fourth switches on the DAC, the I/Q imbalance, the PA and the 41 kHz offset.

The reading is consistent across both constellations and is the useful result. About one decibel is the decoder: normalised min-sum on BG1 is known to sit roughly there behind belief propagation [11], and this is that penalty, measured rather than quoted. A little over four tenths is the entire cost of being blind: timing, fractional and integer CFO, slot identity, channel estimation and noise variance, all together. Under a tenth is every analog impairment at the default 8 dB backoff, which sounds too good until one notices that EVM near 6% is about -24 dB while the cliff sits at 6.7 dB, so the impairments are far below the noise that decides the block. They matter for 256QAM, and Table II is where they show up. The QPSK front-end row reads 0.00 because both thresholds landed in the same bisection bin, so the cost is below the 0.05 dB resolution of the search rather than exactly zero.

Two things about the table matter more than its numbers. It makes a null result legible: contribute a better decoder and the second row moves while the others do not, so the table says in advance how much there is to win. And before the synchroniser bug of Section V was found, this same table read $+7.96$ dB on the QPSK OFDM row against a *smaller* $+3.04$ dB below it, which is impossible, since adding impairments cannot help. The budget caught what a waterfall plot had been quietly absorbing.

B. Against exact theory

The uncoded Gray-QAM error rate on AWGN is computed from the PAM decision regions with no approximation, and the blind chain is held against it. In Fig. 3 the dashed curves are that theory and the measured pre-FEC points ride just above, by 3 to 6% at the two cliffs and 55 to 65% at the top of each sweep, which is the estimation cost showing up exactly where it should (at high SNR the residual is no longer noise but estimator bias, and it stops shrinking). The coded cliffs land near 1.5 dB for QPSK and 6.6 dB for 16QAM, inside 0.2 dB of the bisected thresholds of Table I, which is the grid spacing of the sweep itself and comes from a separate measurement.

At 30 dB with clean analog the chain's EVM averages 3.1 to 3.5% across QPSK through 256QAM over twelve seeds, against a thermal floor of $10^{-30/20} = 3.16\%$, with about 0.2

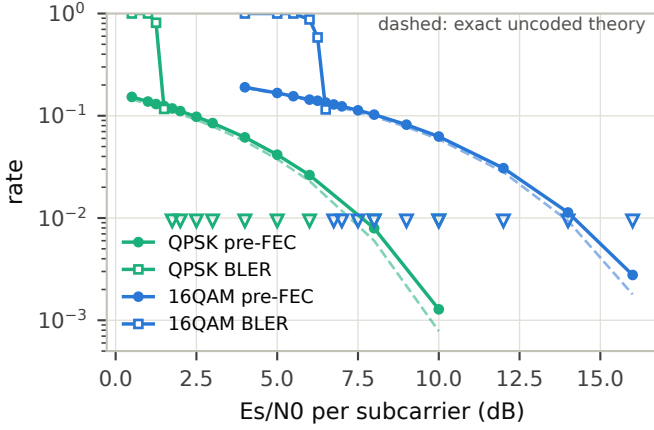


Fig. 3. AWGN waterfalls at rate 1/2: measured pre-FEC BER against the exact uncoded theory (dashed), and coded BLER by CRC24A. Open markers are counting floors. The cliffs sit 1.4 dB from constrained capacity for both constellations.

points of seed-to-seed scatter. The receiver's own contribution is smaller than that scatter, which is the strongest statement the measurement can make: the coded curves show the code and the channel, not the software.

C. The EVM budget

Fig. 5 is the design curve every transmitter engineer draws: EVM against PA input backoff, with the 38.101 limits marked and the 256QAM one shaded, since that band is the only place 256QAM is allowed to live. At the default 8 dB backoff the chain measures 5.6 to 6.2%, inside the limits for QPSK, 16QAM and 64QAM (17.5, 12.5 and 8%). What the curve also shows is that it stops improving: past about 12 dB every curve flattens just under 4% and stays there out to 24 dB, never entering the shaded band. Whatever is left at that point is not compression, and no amount of backoff will touch it.

Table II takes the floor apart by enabling one impairment at a time at the same 45 dB E_s/N_0 , over twelve seeds, where noise and estimation together contribute 1.32%. Removing that baseline in power gives each impairment its own share, and adding the shares back reproduces both totals exactly: 3.86% with the PA off and 6.19% at the default backoff, against 3.86 and 6.19 measured. A budget that closes twice on separate totals is the evidence that the shares are really separate.

Two rows can be checked in closed form. LO leakage at -35 dBc is 1.78% by definition, and the table measures 1.79%. I/Q imbalance is more interesting: 0.2 dB and 1° puts an image 1.44% of the signal at one end, so two ends should give 2.0% if they were independent and 2.9% if their images landed in phase, and the measured 3.12% says the second. That is not an accident of the model. The default gives both ends the same sign, so the two images add in amplitude; setting the receive imbalance to -0.2 dB and -1° drops the share to 0.06%, the two cancelling almost perfectly. Real hardware sits between, but the worst case is the one to design against.

The practical conclusion is the one real transmitters reach. 256QAM needs 3.5% and backoff alone floors at 3.86%, so this transmitter cannot legally carry 256QAM no matter how

TABLE II
TRANSMIT EVM BUDGET, 256QAM AT 45 dB E_s/N_0 , MEAN OF TWELVE SEEDS. EACH IMPAIRMENT IS ENABLED ALONE; ITS SHARE IS THE MEASURED VALUE WITH THE 1.32% BASELINE REMOVED IN POWER. THE LAST TWO ROWS ARE THE CHECK: PREDICTED IS THE SHARES ADDED BACK.

Enabled alone	EVM (%)	share (%)
nothing: noise and estimation	1.32	
I/Q imbalance, Tx and Rx	3.39	3.12
LO leakage at -35 dBc	2.22	1.79
DAC and ADC at 10 bit	1.39	0.46
Rapp PA at 8 dB backoff	5.01	4.84
	measured	predicted
all analog, PA backed right off	3.86	3.86
all analog, PA at 8 dB backoff	6.19	6.19

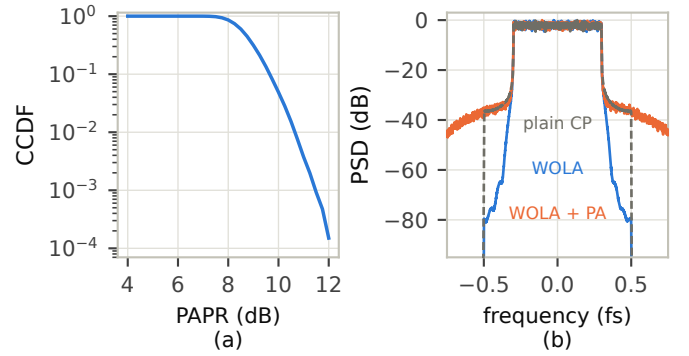


Fig. 4. The waveform classics: (a) PAPR CCDF over 40k symbols, stopped where the counts run out; (b) transmit PSD with a plain CP, with WOLA, and with WOLA through the PA at 8 dB backoff. The cliff at $\pm 0.5f_s$ is the oversampling filter, which the regrowth walks past.

much of the amplifier it wastes. The fix is not backoff but I/Q calibration and digital predistortion, which move the two largest rows of the table instead of trading against them. Both are absent here and both are on the roadmap, and the table says which one to do first.

D. The waveform classics

The PAPR CCDF (Fig. 4a) runs over 40k symbols, so its tail is a measurement and not a sample count: it crosses 10^{-2} at 10.7 dB and 10^{-3} at 11.5 dB, the textbook numbers for hundreds of subcarriers. Windowing (Fig. 4b) buys about 21 dB of out-of-band floor just past the carrier edge and 30 dB further out, taking the plain CP's -31 dBc first skirt to -52 dBc. The PA at 8 dB backoff puts all of it back: the skirt returns to -32 dBc, a decibel from where the unwindowed waveform started, and the regrowth reaches past the oversampling filter's stopband, where both linear waveforms had nothing at all. That is not a defect of the model but the central fact of transmitter design. Digital shaping is free and easily undone by an amplifier a few decibels into compression, which is why adjacent-channel leakage is fought with backoff, crest factor reduction and predistortion rather than better windows.

E. Fading

Over TDL-C at 300 ns delay spread and 70 Hz Doppler, with the 41 kHz CFO and every front-end impairment on, the

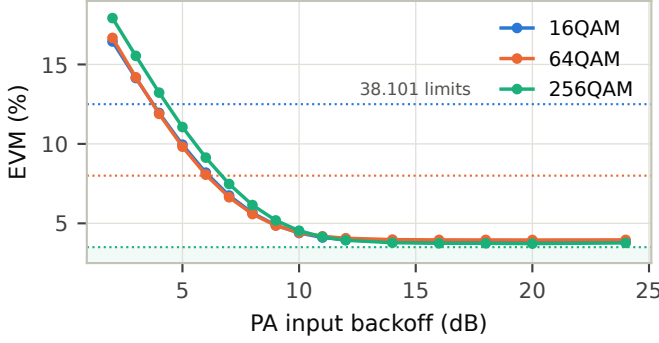


Fig. 5. Transmit EVM against PA input backoff, held against the 38.101 per-order limits (dotted); the 256QAM band is shaded. The default 8 dB backoff clears 64QAM, but every curve flattens under 4% on the I/Q and LO floor of Table II and none reaches the shaded band: 256QAM is a calibration problem, not a backoff problem.

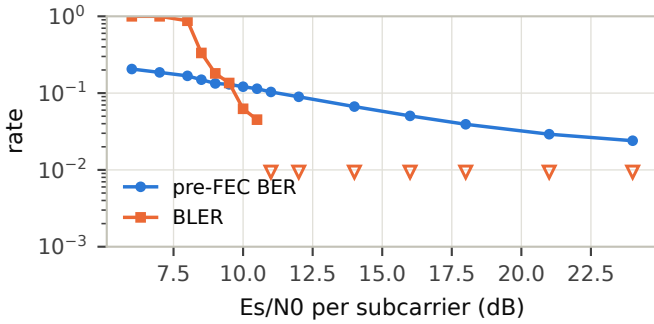


Fig. 6. 16QAM rate-1/2 over TDL-C (300 ns, 70 Hz Doppler) with all impairments: pre-FEC BER and CRC24A BLER against SNR. Open markers are counting floors.

16QAM rate-1/2 BLER walks down as SNR rises (Fig. 6) with the shallower slope fading always writes. The estimator tracks the DM-RS through the fades, the per-RE LLR weighting tells the decoder which subcarriers to distrust, and the CRC24A verdicts count what a scheduler would see. The cliff lands near 9.7 dB, about 3 dB right of the AWGN one, and that modest penalty is the interesting part: flat Rayleigh with no diversity would cost far more, but a code block here is spread over 612 subcarriers and five slots, so the LDPC sees a great many independent fades and recovers most of it. The pre-FEC curve flattens at high SNR for the delay-spread reason of Section IV, not because anything in the receiver has given up.

VII. IMPLEMENTATION NOTES

Everything is numpy; there is no compiled extension. The LDPC machinery is shared with the free-space sibling and vectorises over the lifted graph (an edge is one `np.roll`); the OFDM side is naturally matrix-shaped, a slot being an FFT of a grid. Eight transport blocks (67 k payload bits, five slots of the 20 MHz carrier) take about a second on a laptop core on AWGN; the TDL taps add their convolutions.

The repository ships 26 unit tests, including bit-exact checks of constellation points, CP arithmetic, Gold sequences and the rate-matching interleaver, plus the theory-match and EVM-limit tests. Three encode the synchroniser lesson: one runs a

capture long enough to need nine slots and asserts the timing error has not grown, one asserts the receiver lands on absolute slot 0 across five seeds, and one pins the error to the early side of the prefix. A bug that only appears in long captures and only degrades results gradually is the kind that lives in a simulator for years, and the way to keep it out is to make the length itself part of the test. Every figure regenerates from committed scripts with fixed seeds.

VIII. LIMITATIONS AND ROADMAP

One code block per transport block (the largest BG1 lifting only), rv0 only, single antenna, no transform precoding, no PT-RS (the decision-directed common phase stands in), no sampling-clock offset, and acquisition starts at the CP correlator rather than at SSB/PRACH. The most valuable additions, roughly in order: code-block segmentation with the full lifting table, HARQ redundancy versions, crest-factor reduction and digital predistortion around the PA, 2×2 MIMO with type-1 DM-RS ports, and SSB-based initial access. Table I sizes each of those before anyone writes it, which is the point of keeping it.

IX. CONCLUSION

nrphy makes the whiteboard OFDM diagram executable and holds it to the 5G standard's bits: 38.212 coding and 38.211 modulation exact, TR 38.901 fading, a receiver that synchronises and estimates blind, EVM inside the 38.101 limits, and coded cliffs 1.4 dB from constrained capacity at rate 1/2. More useful than any of those single numbers is the decomposition behind them, one decibel of decoder, four tenths of blind receiver, a sliver of front end, and an EVM budget whose parts add up. With its fibre and free-space siblings it completes a trilogy of open, readable physical layers, offered as a base that anyone can verify, break and extend.

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